# Serverless Overview

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## What's serverless?

- Serverless is a new paradigm in which the developers don't have to manage servers anymore…
- They just deploy code
- They just deploy… functions !
- Initially... Serverless == FaaS (Function as a Service)
- Serverless was pioneered by AWS Lambda but now also includes anything that's managed: "databases, messaging, storage, etc."
- **Serverless does not mean there are no servers…**
  it means you just don't manage / provision / see them

# Why AWS Lambda

**Amazon EC2**

- Virtual Servers in the Cloud
- Limited by RAM and CPU
- Continuously running
- Scaling means intervention to add / remove servers

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**Amazon Lambda**

- Virtual **functions** – no servers to manage!
- Limited by time - **short executions**
- Run **on-demand**
- Scaling is **automated**!

# Benefits of AWS Lambda

- Easy Pricing:
  - Pay per request and compute time
  - Free tier of 1,000,000 AWS Lambda requests and 400,000 GBs of compute time
- Integrated with the whole AWS suite of services
- Integrated with many programming languages
- Easy monitoring through AWS CloudWatch
- Easy to get more resources per functions (up to 10GB of RAM!)
- Increasing RAM will also improve CPU and network!

# AWS Lambda language support

- Node.js (JavaScript)
- Python
- Java
- C# (.NET Core) / Powershell
- Ruby
- Custom Runtime API (community supported, example Rust or Golang)

- Lambda Container Image
  - The container image must implement the Lambda Runtime API
  - ECS / Fargate is preferred for running arbitrary Docker images

# AWS Lambda Integrations
# Main ones

API Gateway

Kinesis

DynamoDB

S3

CloudFront

CloudWatch Events
EventBridge

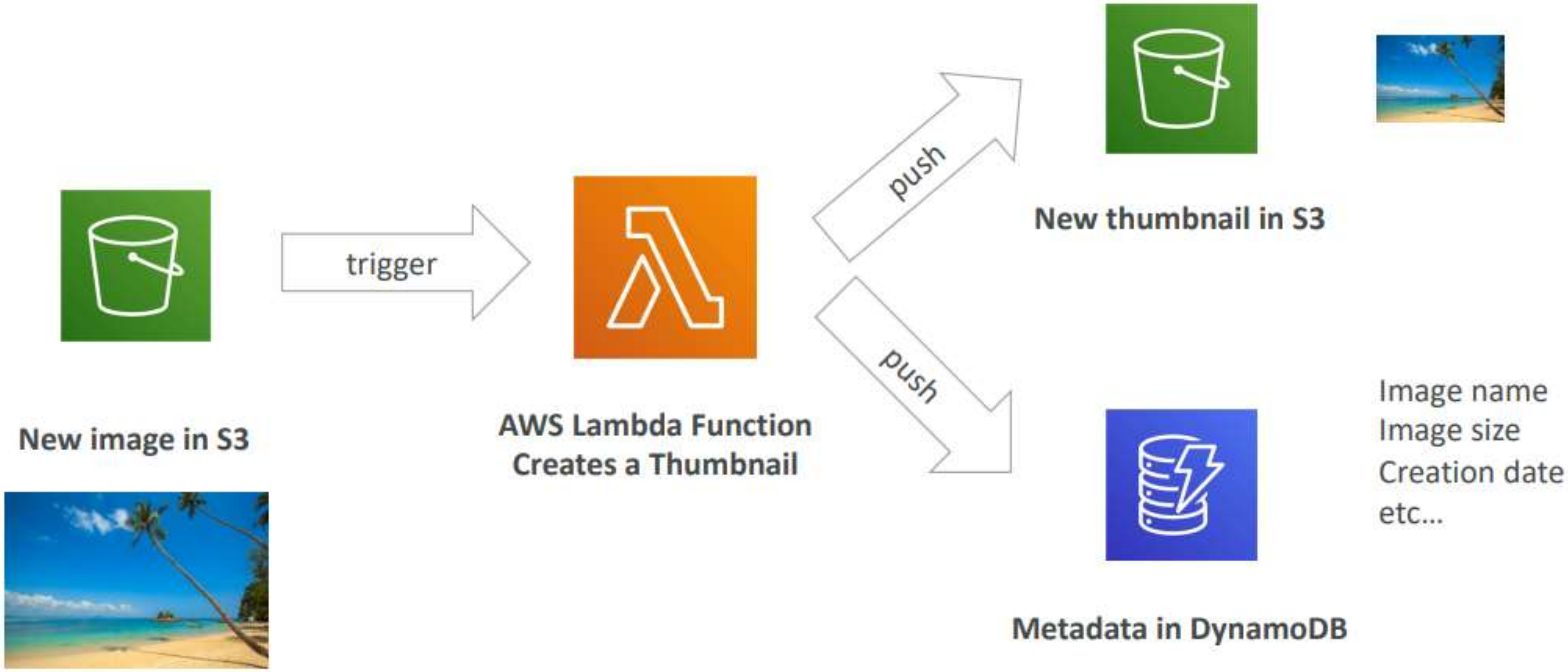
CloudWatch Logs

SNS

SQS

Cognito

# Example: Serverless Thumbnail creation



# Example: Serverless CRON Job


**CloudWatch Events**
**EventBridge**

**AWS Lambda Function**
**Perform a task**

# AWS Lambda Pricing: example

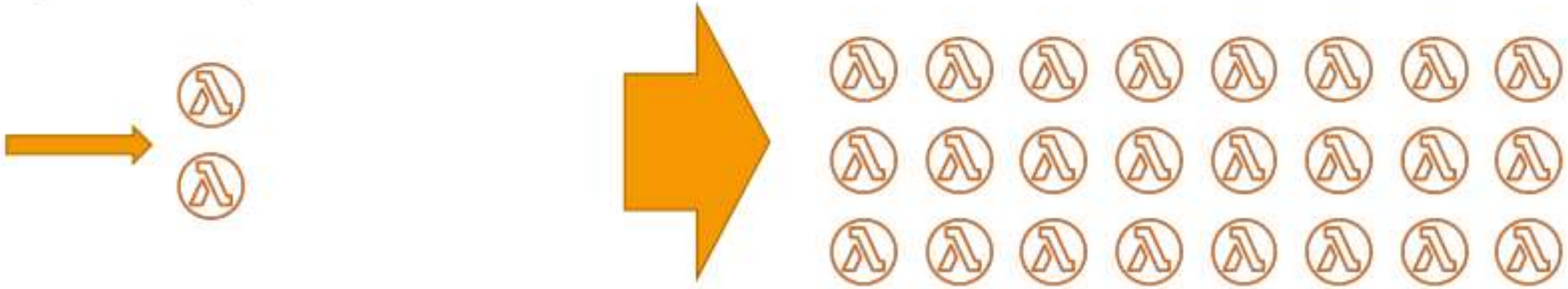
- You can find overall pricing information here: https://aws.amazon.com/lambda/pricing/
- Pay per **calls**:
  - First 1,000,000 requests are free
  - $0.20 per 1 million requests thereafter ($0.0000002 per request)
- Pay per **duration**: (in increment of 1 ms)
  - 400,000 GB-seconds of compute time per month for FREE
  - == 400,000 seconds if function is 1GB RAM
  - == 3,200,000 seconds if function is 128 MB RAM
  - After that $1.00 for 600,000 GB-seconds
- It is usually very cheap to run AWS Lambda so it's very popular

# AWS Lambda Limits to Know - **per region**

- **Execution:**
  - Memory allocation: 128 MB – 10GB (1 MB increments)
  - Maximum execution time: 900 seconds (15 minutes)
  - Environment variables (4 KB)
  - Disk capacity in the "function container" (in /tmp): 512 MB to 10GB
  - Concurrency executions: 1000 (can be increased)
- **Deployment:**
  - Lambda function deployment size (compressed .zip): 50 MB
  - Size of uncompressed deployment (code + dependencies): 250 MB
  - Can use the /tmp directory to load other files at startup
  - Size of environment variables: 4 KB

# Lambda Concurrency and Throttling

- Concurrency limit: up to 1000 concurrent executions



- Can set a "**reserved concurrency**" at the function level (=limit)
- Each invocation over the concurrency limit will trigger a "Throttle"
- Throttle behavior:
  - If synchronous invocation => return ThrottleError - 429
  - If asynchronous invocation => retry automatically and then go to DLQ
- If you need a higher limit, open a support ticket

# Cold Starts & Provisioned Concurrency

- Cold Start:
  - New instance => code is loaded and code outside the handler run (init)
  - If the init is large (code, dependencies, SDK…) this process can take some time.
  - First request served by new instances has higher latency than the rest
- Provisioned Concurrency:
  - Concurrency is allocated before the function is invoked (in advance)
  - So the cold start never happens and all invocations have low latency
  - Application Auto Scaling can manage concurrency (schedule or target utilization)
- Note:
  - Note: cold starts in VPC have been dramatically reduced in Oct & Nov 2019
  - https://aws.amazon.com/blogs/compute/announcing-improved-vpc-networking-for-aws-lambda-functions/

# Customization At The Edge

- Many modern applications execute some form of the logic at the edge
- **Edge Function:**
  - A code that you write and attach to CloudFront distributions
  - Runs close to your users to minimize latency
- CloudFront provides two types: **CloudFront Functions** & **Lambda@Edge**
- You don't have to manage any servers, deployed globally

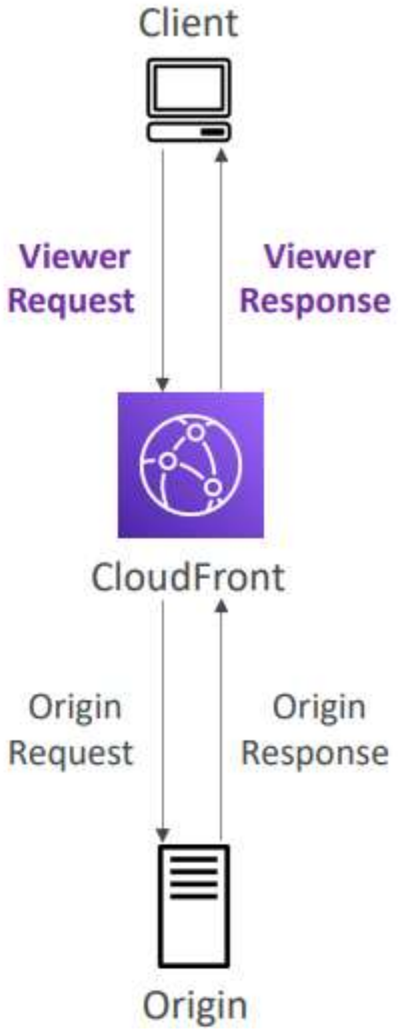
- Use case: customize the CDN content
- Pay only for what you use
- Fully serverless

# CloudFront Functions & Lambda@Edge Use Cases

- Website Security and Privacy
- Dynamic Web Application at the Edge
- Search Engine Optimization (SEO)
- Intelligently Route Across Origins and Data Centers
- Bot Mitigation at the Edge
- Real-time Image Transformation
- A/B Testing
- User Authentication and Authorization
- User Prioritization
- User Tracking and Analytics

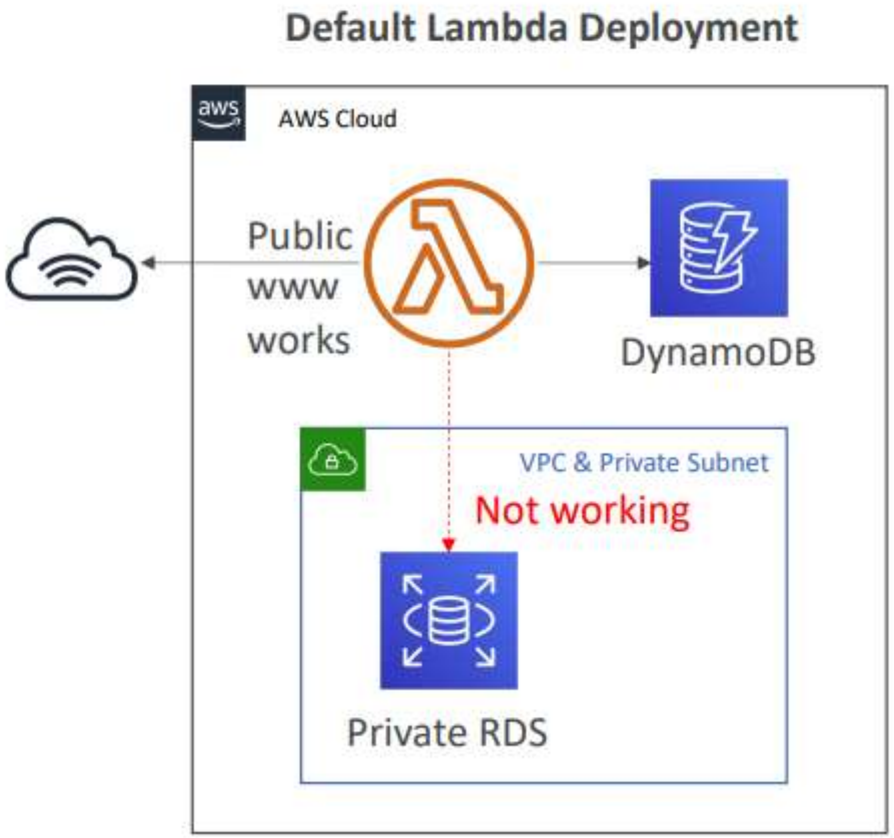
# CloudFront Functions

- Lightweight functions written in JavaScript
- For high-scale, latency-sensitive CDN customizations
- Sub-ms startup times, **millions of requests/second**
- Used to change Viewer requests and responses:
  - **Viewer Request:** after CloudFront receives a request from a viewer
  - **Viewer Response:** before CloudFront forwards the response to the viewer
- Native feature of CloudFront (manage code entirely within CloudFront)



# Lambda by default

- By default, your Lambda function is launched outside your own VPC (in an AWS-owned VPC)
- Therefore, it cannot access resources in your VPC (RDS, ElastiCache, internal ELB…)

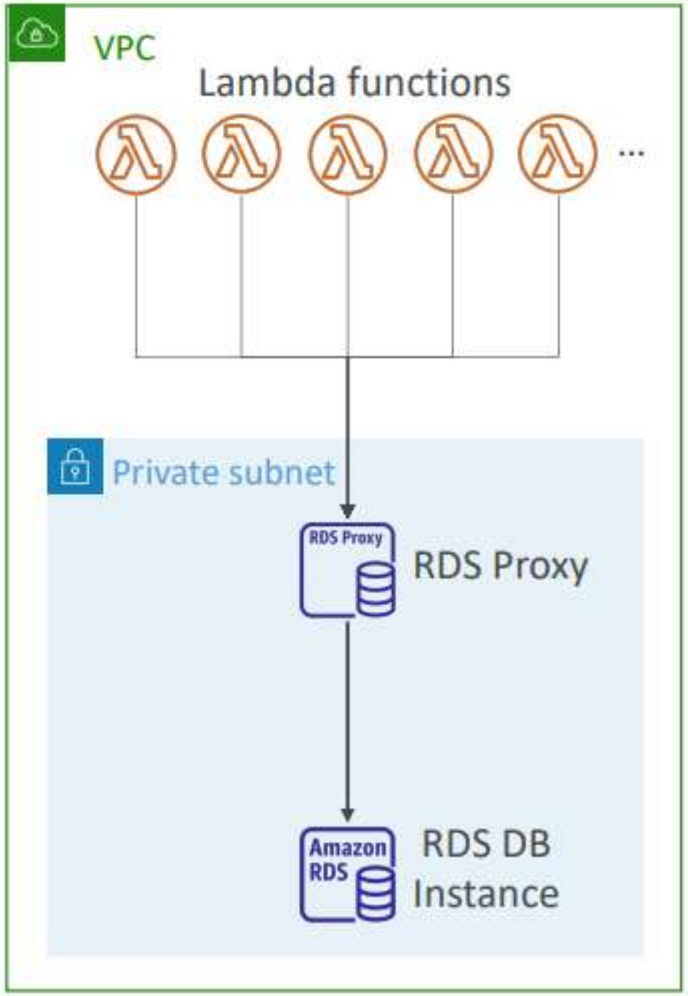


# Lambda in VPC

- You must define the VPC ID, the Subnets and the Security Groups
- Lambda will create an ENI (Elastic Network Interface) in your subnets

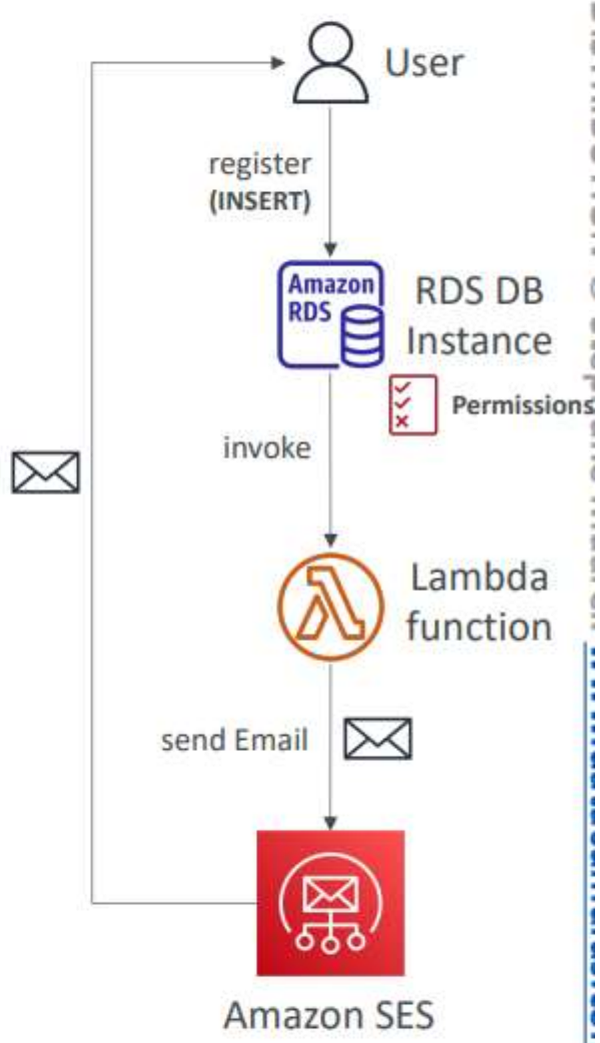
# Lambda with RDS Proxy

- If Lambda functions directly access your database, they may open too many connections under high load
- RDS Proxy
  - Improve scalability by pooling and sharing DB connections
  - Improve availability by reducing by 66% the failover time and preserving connections
  - Improve security by enforcing IAM authentication and storing credentials in Secrets Manager
- **The Lambda function must be deployed in your VPC, <u>because RDS Proxy is never publicly accessible</u>**



# Invoking Lambda from RDS & Aurora

- Invoke Lambda functions from within your DB instance
- Allows you to process **data events** from within a database
- Supported for **RDS for PostgreSQL and Aurora MySQL**
- **Must allow outbound traffic to your Lambda function** from within your DB instance (Public, NAT GW, VPC Endpoints)
- **DB instance must have the required permissions to invoke the Lambda function** (Lambda Resource-based Policy & IAM Policy)



# Amazon DynamoDB

- Fully managed, highly available with replication across multiple AZs
- NoSQL database - not a relational database - with transaction support
- Scales to massive workloads, distributed database
- Millions of requests per seconds, trillions of row, 100s of TB of storage
- Fast and consistent in performance (single-digit millisecond)
- Integrated with IAM for security, authorization and administration
- Low cost and auto-scaling capabilities
- No maintenance or patching, always available
- Standard & Infrequent Access (IA) Table Class

# DynamoDB - Basics

- DynamoDB is made of **Tables**
- Each table has a **Primary Key** (must be decided at creation time)
- Each table can have an infinite number of items (= rows)
- Each item has **attributes** (can be added over time – can be null)
- Maximum size of an item is **400KB**
- Data types supported are:
  - **Scalar Types** – String, Number, Binary, Boolean, Null
  - **Document Types** – List, Map
  - **Set Types** – String Set, Number Set, Binary Set
- <u>Therefore, in DynamoDB you can rapidly evolve schemas</u>

# DynamoDB – Read/Write Capacity Modes

- Control how you manage your table's capacity (read/write throughput)

- **Provisioned Mode (default)**
  - You specify the number of reads/writes per second
  - You need to plan capacity beforehand
  - Pay for <u>provisioned</u> Read Capacity Units (RCU) & Write Capacity Units (WCU)
  - Possibility to add <u>auto-scaling</u> mode for RCU & WCU

- **On-Demand Mode**
  - Read/writes automatically scale up/down with your workloads
  - No capacity planning needed
  - Pay for what you use, more expensive ($$$)
  - Great for <u>unpredictable</u> workloads, <u>steep sudden spikes</u>

# DynamoDB – Backups for disaster recovery

- Continuous backups using point-in-time recovery (PITR)
  - Optionally enabled for the last 35 days
  - Point-in-time recovery to any time within the backup window
  - The recovery process creates a new table

- On-demand backups
  - Full backups for long-term retention, until explicitely deleted
  - Doesn't affect performance or latency
  - Can be configured and managed in AWS Backup (enables cross-region copy)
  - The recovery process creates a new table

# DynamoDB – Integration with Amazon S3

- Export to S3 (must enable PITR)
  - Works for any point of time in the last 35 days
  - Doesn't affect the read capacity of your table
  - Perform data analysis on top of DynamoDB
  - Retain snapshots for auditing
  - ETL on top of S3 data before importing back into DynamoDB
  - Export in DynamoDB JSON or ION format


- Import from S3
  - Import CSV, DynamoDB JSON or ION format
  - Doesn't consume any write capacity
  - Creates a new table
  - Import errors are logged in CloudWatch Logs

# AWS API Gateway

- AWS Lambda + API Gateway: No infrastructure to manage
- Support for the WebSocket Protocol
- Handle API versioning (v1, v2…)
- Handle different environments (dev, test, prod…)
- Handle security (Authentication and Authorization)
- Create API keys, handle request throttling
- Swagger / Open API import to quickly define APIs
- Transform and validate requests and responses
- Generate SDK and API specifications
- Cache API responses

# Amazon Cognito

- Give users an identity to interact with our web or mobile application
- **Cognito User Pools:**
  - Sign in functionality for app users
  - Integrate with API Gateway & Application Load Balancer

- **Cognito Identity Pools (Federated Identity):**
  - Provide AWS credentials to users so they can access AWS resources directly
  - Integrate with Cognito User Pools as an identity provider

- **Cognito vs IAM:** "hundreds of users", "mobile users", "authenticate with SAML"

# API Gateway - Endpoint Types

- **Edge-Optimized (default):** For global clients
  - Requests are routed through the CloudFront Edge locations (improves latency)
  - The API Gateway still lives in only one region
- **Regional:**
  - For clients within the same region
  - Could manually combine with CloudFront (more control over the caching strategies and the distribution)
- **Private:**
  - Can only be accessed from your VPC using an interface VPC endpoint (ENI)
  - Use a resource policy to define access

# Cognito Identity Pools – Diagram